

Model-completed bathymetry and a shoal-hazard field for the Churchill trade corridor, validated on soundings the model never saw

Emilio Girard · SeabedNet, Montréal · girardemilio3@gmail.com

Interactive atlas: girardemilio3-svg.github.io/churchill-corridor-atlas · Code and results: github.com/girardemilio3-svg/seabednet-validation

PLANNING PRIOR — NOT FOR NAVIGATION

Abstract. Canada, Manitoba and Saskatchewan have committed C\$262.5M to reopen the Churchill trade corridor through Hudson Bay and Hudson Strait, water in which the Canadian Hydrographic Service (CHS) reports 15.8% of Arctic waters and 44.7% of key routes surveyed to modern standards. Under the 2,327 km Churchill–Atlantic route itself, 17% of the seabed carries a published sounding; the median survey year of those soundings is 1974. We train a heteroscedastic masked-completion network on the public CHS NONNA archive and score it on two sets of depths it never saw. (1) 109,044 research-cruise multibeam cells on the shelf (50–400 m) at locations where NONNA has no sounding: mean absolute error 4.9 m, against 13.4 m for the nearest published sounding and 5.5 m for the gravity-derived prior. (2) A temporal split using CHS's own dated Survey Index: a model trained only on soundings from before 2016 predicts the 6.46 million post-2016 soundings in the corridor to 13.3 m (18.8 m nearest sounding, 18.1 m gravity), with 74% of errors inside the model's 1σ . Because ships ground on the shallowest point and not the mean, we further train a hazard model on NONNA-10/100 pairs to predict the minimum depth within 500 m (error 2.7 m on held-out tiles versus 11.0 m if the archive depth is taken as the shoal) and hindcast 7 Transportation Safety Board groundings using only pre-incident soundings: the strike site falls in the top 10% of hazard among apparently safe water in 5 of 7 cases. We publish a calibrated uncertainty field, a σ -ranked survey plan (219 ship-days, C\$40M), a sealed forecast of

1,314 predicted depths (SHA-256 on record), and the full code and validation cells. Limitations are stated in §8.

1. Problem

Hydrographic adequacy in the Canadian Arctic is a documented gap: the CHS, reporting through the IHO in December 2025, put 15.8% of Canadian Arctic waters and 44.7% of key routes at adequate survey standard [1]; in 2016 its Hydrographer-in-Charge for the Arctic put full charting "more than a decade" away [2]. Groundings on uncharted or poorly charted shoals recur — Hanseatic (1996), Clipper Adventurer (2010), Akademik Ioffe (2018), Thamesborg (2025) — and the TSB reports in each case identify the survey state of the water as a contributing factor [3–6].

The federal–provincial Churchill Plus commitment (C\$262.5M, February 2026) funds port and rail, not charting. The question this report addresses is narrow: given only the public archive, how much of the corridor's seabed can be inferred with stated, tested uncertainty, and where should a survey ship go first? A second question follows from the first and is, to our knowledge, unaddressed in the completion literature: a completion model predicts mean depth, but a keel meets the shallowest point. We treat that quantity as a separate target.

2. Data

Archive. CHS NONNA-100 (100 m) and NONNA-10 (10 m) non-navigational bathymetry, Open Government Licence – Canada, retrieved on 29 August 2026 through the CHS WCS endpoint. 437 national 100 m blocks ($\approx 2,000 \times 2,000$ cells) and 3,601 10 m tiles; corridor subset of 94 blocks covering Hudson Bay, Hudson Strait, the Labrador approaches and Franklin Strait. NONNA is the *published* archive; CHS holds surveys not yet released to it. Throughout, "no published sounding" is the only claim made; "unsurveyed" is never inferred.

Physics prior. SRTM15+ V2.7 gravity-derived bathymetry (Scripps) [7], resampled to a 295 MB Canada raster; used as an input channel and as the always-reported baseline.

Independent multibeam. Gridded multibeam from the GMRT synthesis (Lamont-Doherty) [8], topo-mask layer, which returns measured cells only. In the corridor bounding box these derive from NCEI-archived research cruises (CCGS Amundsen 2003–2013, USCGC Healy, R/V Knorr, R/V Neil Armstrong, R/V Maria S. Merian; 36 surveys). Independence from NONNA is by provenance: research-cruise data are not CHS holdings.

Survey dates. The CHS Survey Index (DFO EGIS), 7,078 polygons with survey start/end and CATZOC grade, 1832–2016 [9]. Rasterized onto every block, it dates each sounding to the latest indexed survey covering it; a sounding outside every polygon is, by construction, post-2016. Nationally, 46.8% of NONNA-100 soundings and 46.5% of NONNA-10 soundings are post-index; in the corridor, 19,471,411 of 34,281,587 (56.8%).

Groundings. Positions, dates and drafts from TSB reports M96H0016, M10H0006, M12H0012, M14C0219, M18C0225 and the open M25C0241 occurrence [3–6]; the Thamesborg position is from AIS and press reporting and is flagged as such.

3. Method

3.1 Completion model (mean depth and uncertainty)

A hierarchical ConvNeXt-style U-Net with self-attention at the two coarsest scales, FiLM-conditioned on a learned resolution embedding (10 m / 100 m). Inputs per 256×256 patch: visible depth, visibility mask, gravity prior. All depth channels are normalized by the gravity patch statistics (mean, standard deviation floored at 5 m), so normalization is continuous across windows and the prediction is a residual over physics. Two heads: μ and $\log \sigma^2$, trained with the heteroscedastic Gaussian negative log-likelihood on hidden cells plus an L1 term on visible cells. Masks mix harvested real NONNA gap patterns, random blocks and half-plane survey edges. Configurations: tiny (6.8M parameters) and small (34.8M; widths 96–768, depths 2/3/6/3). AdamW 2×10^{-4} , cosine schedule, batch 16, bf16; 4,000 steps. Inference: overlapping windows (stride 128) with Hann blending; windows require ≥ 400 visible cells; predicted land ($\mu > -4$ m) is dropped; inference is published only within 6 km of a sounding.

3.2 Hazard model (shallowest point)

Where NONNA-10 and NONNA-100 overlap, the 10 m grid gives the sub-cell extreme statistics the 100 m grid hides. For each 100 m cell we compute the shallowest 10 m sounding within a 500 m radius (a 101×101 maximum filter on negative depths, requiring ≥50% coverage of the disc). 3,601 tiles mosaicked 4×4 give 475 groups of 800×800 cells. Along the Churchill route the shallowest point sits a median 6 m above the 100 m mean. The same backbone is trained to predict this extreme and its σ from the masked 100 m field and gravity, everywhere — the extreme is unknown even where a 100 m sounding exists. The hazard for a draft d is

$$P(\text{shoal} < d) = 1 - \Phi((-d - \mu_s) / \sigma_s)$$

Groups within 0.4° of any of the six grounding sites and the three geographic holdouts are excluded from training. Configurations: tiny (1,500 steps) and small (3,000 steps).

3.3 Validation design

Test 1 (independent). Cells with a GMRT measured depth, no NONNA sounding, and a model fill within 60 cells (6 km) of a sounding; baselines are the nearest published sounding (via distance transform) and the gravity prior; stratified by distance and depth. **Test 2 (temporal).** The model is retrained with every post-index sounding blanked from the corpus (100 m and 10 m); it is then given the pre-2016 soundings of each corridor block and scored on the post-2016 ones. **σ calibration.** An isotonic map from raw σ to the 68th and 95th percentiles of $|\text{error}|$ is fitted on Test 1 shelf cells and applied everywhere. **Hindcast.** For each grounding, the hazard model receives only soundings dated before the incident (all soundings for Thamesborg, since NONNA carries no post-2016 dates); the strike cell's $P(\text{shoal} < \text{draft})$, allowing 500 m of position slop, is ranked among "apparently safe" cells — mean-model depth deeper than twice the draft — within 25 km. A rank at or above the 90th percentile is counted as a landing.

4. Results: completion

4.1 Test 1 — research-cruise multibeam the archive does not contain

DISTANCE TO NEAREST SOUNDING	N CELLS	SEABEDNET (M)	NEAREST SOUNDING (M)	GRAVITY PRIOR (M)
0–0.5 km	37,313	4.8	8.5	5.9
0.5–1 km	31,657	4.6	12.9	5.5
1–2 km	27,743	5.0	17.5	4.8
2–4 km	12,151	5.5	20.8	5.8
<i>trend + natural-neighbour residual</i>			11.2	
<i>trend + inverse-distance residual</i>			8.7	
<i>GEBCO reference (not independent, §4.5)</i>			3.0	
all shelf 50–400 m	109,044	4.9	13.4	5.5

Table 1. Mean absolute error on independent multibeam cells, shelf water 50–400 m, by distance from the nearest published sounding. 386,685 cells over 17 blocks in total, of which 109,044 on the shelf.

DEPTH BAND	N CELLS	SEABEDNET (M)	GRAVITY PRIOR (M)
coastal < 50 m (excluded, see §4.4)	5,858	21.4	11.8
shelf 50–400 m	109,044	4.9	5.5
slope 400–1000 m	5,849	29.5	6.4
basin > 1000 m	265,934	10.9	6.3

Table 2. The same test by depth. Below 400 m the gravity prior is the better estimator; the published completion therefore defers to it there (300–400 m blended).

On the shelf the model reduces error by 64% relative to the archive's own nearest sounding and the advantage grows with distance (5.5 vs 20.8 m at 2–4 km). Against the gravity prior the margin on these cells is small (4.9 vs 5.5 m): research cruises transit deep channels where gravity inversion already works, and the glacial shelf where it fails is where NONNA is dense and therefore absent from this test. Test 2 addresses that.

4.2 Test 2 — pre-2016 model, post-2016 soundings

DISTANCE TO NEAREST PRE-2016 SOUNDING	N CELLS	SEABEDNET (M)	NEAREST SOUNDING (M)	GRAVITY (M)	BIAS (M)	ERR ≤ 1σ
0–0.5 km	998,636	7.1	5.6	17.8	-0.3	65%
0.5–1 km	910,868	10.9	11.8	17.9	+0.6	72%
1–2 km	1,642,606	13.3	17.1	17.5	-1.1	75%
2–4 km	2,543,006	15.6	25.4	18.2	+1.1	77%
4–8 km	365,496	20.1	34.9	20.7	+3.2	75%
<i>trend + natural- neighbour residual</i>			16.8			
<i>trend + inverse- distance residual</i>			16.2			
<i>GEBCO reference (not independent, §4.5)</i>			6.2			
all	6,460,612	13.3	18.8	18.1	+0.4	74%

Table 3. Temporal holdout, corridor, 34.8M-parameter model. Positive bias = model shallower than the sounding. The 6.8M model scored 13.5 m overall on the same cells; model capacity is not the limiting factor.

DEPTH BAND	N CELLS	SEABEDNET (M)	NEAREST (M)	GRAVITY (M)	BIAS (M)
0–20 m	118,374	15.3	21.1	11.1	-9.8
20–50 m	311,616	17.9	21.6	13.8	-11.0
50–100 m	672,069	13.9	17.4	15.3	-3.9
100–200 m	1,741,821	8.4	12.2	9.7	+0.3
200–400 m	2,346,496	11.9	17.2	17.2	+2.0
400–9999 m	1,254,668	20.8	31.1	34.7	+3.4

Table 4. Temporal holdout by depth. In water shallower than 50 m the mean-depth model reads 10–11 m too deep — the direction that matters for a keel, and the motivation for §5.

4.3 Calibration and deferral

Raw σ ranked error correctly (Spearman-like correlation 0.30 on Test 1 shelf cells) but was under-confident: 48% of errors inside 1σ where 68% is expected, 74% inside 2σ . The isotonic map (mean factor 1.9 $\times$) restores 68% / 95% coverage; the temporal test, scored with the raw σ of a differently trained model, independently shows 74% inside 1σ . All uncertainty on the atlas is the calibrated one. Inferred cells deeper than 400 m carry the gravity prior (Table 2).

RAW Σ DECILE (M)	N	ERR P50 (M)	ERR P90 (M)	INSIDE 1σ
0.3–2.1	38,697	5.0	14.1	13%
2.1–3.0	38,749	4.1	12.3	32%
3.0–3.8	38,720	4.2	12.4	42%
3.8–4.9	38,719	3.7	11.7	56%
4.9–5.9	38,894	4.2	11.3	60%
5.9–7.2	38,908	4.8	12.2	64%
7.2–10.3	38,716	5.3	15.5	69%
10.3–17.8	38,808	6.9	21.7	76%
17.8–29.7	38,751	9.1	29.2	85%
29.7–135.6	38,705	19.3	76.1	79%

Table 5. Raw- σ calibration curve on all Test 1 cells.

4.4 Excluded cells

5,858 Test 1 cells shallower than 50 m, at a median 131 m from a sounding and all at the shoreline of three blocks, disagree with every source: the model reads -20 m, the nearest CHS sounding -39 m, and gravity +9 m relative to GMRT. We attribute this to GMRT's coastal grid rather than measured swath and exclude the band from the headline pending inspection; it is reported here rather than removed.

4.5 Stronger baselines, GEBCO, and gravity leakage

Nearest-sounding is a floor. The classical gap fill is a gravity trend plus interpolated residuals (sounding minus SRTM15+ at the input soundings),

interpolated by Delaunay natural-neighbour-style linear interpolation or by inverse-distance-squared over the 12 nearest residuals. On Test 1 these score 11.2 and 8.7 m against the model's 4.9 m; on Test 2, 16.8 and 16.2 m against 13.3 m (Tables 1 and 3, italic rows). GEBCO (NCEI global mosaic, GEBCO 2024/25, 15 arc-second) is reported as the chart-world reference and scores 3.0 m (Test 1) and 6.2 m (Test 2) on the 100 m grid; it is not an independent baseline because it ingests NONNA-100 through IBCAO v5 (the Arctic Seabed 2030 compilation) and the same NCEI cruise archive; its lower error on both tests is the residual of resampling data it already contains, not a prediction, and where it has no source soundings it reduces to SRTM15+, the gravity row. **Gravity leakage.** SRTM15+ V2.7 (released April 2025) inherits the cumulative NCEI multibeam archive of its point releases. Its error on the Test 1 cells is 5.5 m against 15.2 m on 282,193 ordinary NONNA-sounded shelf cells in the same blocks; the prior has almost certainly seen the cruise data. This favours the gravity baseline, which the model still edges on Test 1, and the model, which takes gravity as an input, inherits part of it. Test 2 is therefore the clean comparison: those NONNA shelf soundings are below SRTM15+'s resolution, and there gravity scores 18.1 m against the model's 13.3 m.

5. Results: hazard field and grounding hindcast

On held-out tile groups (around the grounding sites and the geographic holdouts), the 34.8M hazard model predicts the shallowest point within 500 m to 2.7 m mean absolute error where a 100 m sounding exists (11.0 m if that sounding is taken as the shoal; 12.3 m gravity) and 8.2 m where no sounding exists (11.0 m nearest field); 81% of errors inside 1σ , 96% inside 2σ . The 6.8M model: 4.0 / 9.4 m.

THE HAZARD FIELD – P(shallowest point within 500 m < 10.5 m draft), every cell of the corridor

black = safe · grey = no inference · yellow = >50% · blue line = route · red rings = route km the mean map calls safe (>21 m) but the hazard field flags (P>5%): 26 km of 1762

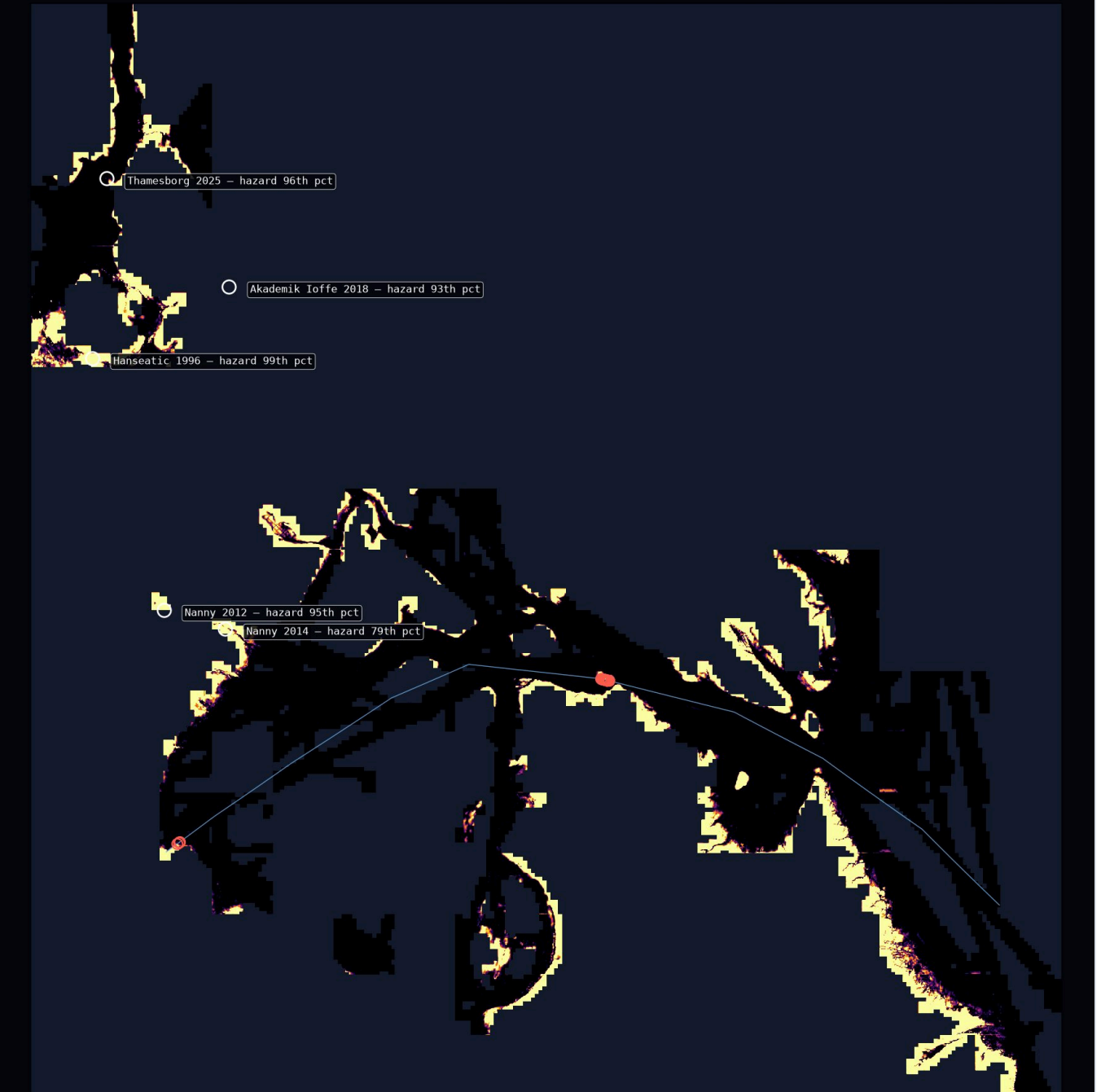


Figure 1. P(shallowest point within 500 m < 10.5 m) for every corridor cell. Blue: the Churchill route; red rings: the 26 route-km (of 1762 km deeper than 21 m on the mean map) where the hazard exceeds 5%; white rings: grounding sites with their hindcast percentile.

GROUNDING	PRE-INCIDENT SOUNDINGS IN BLOCK	KM TO NEAREST	P(SHOAL < DRAFT) AT SITE	HAZARD PERCENTILE	MEAN-Σ PERCENTILE	NEAREST-SOUNDING PERCENTILE
Thamesborg 2025-09-06	1,000,495	0.8	27%	96	99	90

GROUNDING	PRE- INCIDENT SOUNDINGS IN BLOCK	KM TO NEAREST	P(SHOAL < DRAFT) AT SITE	HAZARD PERCENTILE	MEAN-Σ PERCENTILE	NEAREST- SOUNDING PERCENTILE
· TSB M25C0241						
Akademik Ioffe 2018-08-24 · TSB M18C0225	165,454	9.4	26%	93	77	90
Clipper Adventurer 2010-08-27 · TSB M10H0006	16,457	12.4	19%	84	11	22
Hanseatic 1996-08-29 · TSB M96H0016	148,672	0.3	94%	99	18	91
Nanny 2012 2012-10-25 · TSB M12H0012	12,501	6.3	95%	95	74	98
Nanny 2014 2014-10-14 · TSB M14C0219	36,222	0.0	62%	79	98	82
Mokami 2000 2000-10-31 · TSB M00N0098	208,248	0.0	82%	95	11	94

Table 6. Hindcast. Percentiles are among apparently-safe cells within 25 km; higher is more dangerous. Landings (≥ 90 th): Thamesborg (96th), Akademik Ioffe (93rd), Hanseatic (99th), Nanny 2012 (95th), Mokami 2000 (95th). Akademik Ioffe and Clipper Adventurer lie 9–12 km from any pre-incident sounding, beyond the 6 km publication envelope; their scores are reported at a 12 km envelope. The two Nanny cases and Mokami 2000 (Bridges Passage, Labrador; buoy off station, datum mismatch) were navigation errors in known narrows and serve as controls; the hazard model's training did not exclude the Mokami tiles. The TSB

counts 74 Arctic grounding/bottom-contact occurrences for 2000–2018, but publishes no occurrence-level positions; these 7 are every incident with a sourceable position. Four more are known by place name only (Nanny 2010, Mokami 2010, Dorsch 2012, Rosaire A. Desgagnés 2025) and are not scored.

The hazard percentile separates the four uncharted-shoal groundings from the baselines in a way neither the mean model's σ nor the nearest-sounding depth does: for Hanseatic the mean model's σ ranks the site at the 18th percentile (the area was densely sounded, so the mean model was confident) while the hazard field ranks it 99th. For Thamesborg the two agree (96th hazard, 99th σ), which is consistent with the court's finding that the ship was routed through CATZOC C water 1.7 km beyond the modern multibeam swath.

5.1 Blind inputs and sensitivity

Table 6 lets the model see every published sounding around a site at inference; NONNA carries no dates after 2016, so a post-incident survey could be among them. Table 6b removes every sounding within 10 km of each site from the input as well (the hazard model was trained with all tile groups within 0.4° of the sites excluded); a 25 km hole is also shown, at which the site lies outside the 12 km inference envelope and the model returns no estimate. The two denominator choices are then varied: comparison window (10/25/50 km) and the apparently-safe filter (1.5×, 2×, 3× draft on the mean map). Skill: by construction 10% of apparently-safe cells lie above the 90th percentile, so with the 10 km hole the 4-of-7 result has binomial $p = 0.0027$ (all six) and 3 of 4, $p = 0.0037$, on the four uncharted-shoal groundings; with all soundings visible, 5 of 7 ($p = 0.0002$). Precision cannot be bounded from six incidents; corridor-wide 18% of cells exceed $P = 5\%$ and 26 of 1762 apparently-safe route-km are flagged.

GROUNDING	ALL SOUNDINGS	10 KM HOLE	25 KM HOLE	10 KM HOLE · 10 KM WINDOW	10 KM HOLE · 50 KM WINDOW	10 KM HOLE · SAFE 1.5×	10 KM HOLE · SAFE 3×
Thamesborg	96	93	—	91	90	92	94
Akademik Ioffe	93	94	—	74	98	93	94
Clipper Adventurer	84	84	—	99	83	83	84

GROUNDING	ALL SOUNDINGS	10 KM HOLE	25 KM HOLE	10 KM HOLE · 10 KM WINDOW	10 KM HOLE · 50 KM WINDOW	10 KM HOLE · SAFE 1.5×	10 KM HOLE · SAFE 3×
Hanseatic	99	95	—	86	95	86	99
Nanny 2012	95	96	—	100	97	95	98
Nanny 2014	79	60	—	33	70	53	73
Mokami 2000	95	67	—	81	80	63	76

Table 6b. Hazard percentile of the strike site under input holes and denominator choices (window / safe-filter). 10 km hole with the 25 km window and 2× draft is the reference blind number; at 25 km the model has no input within its envelope and abstains.

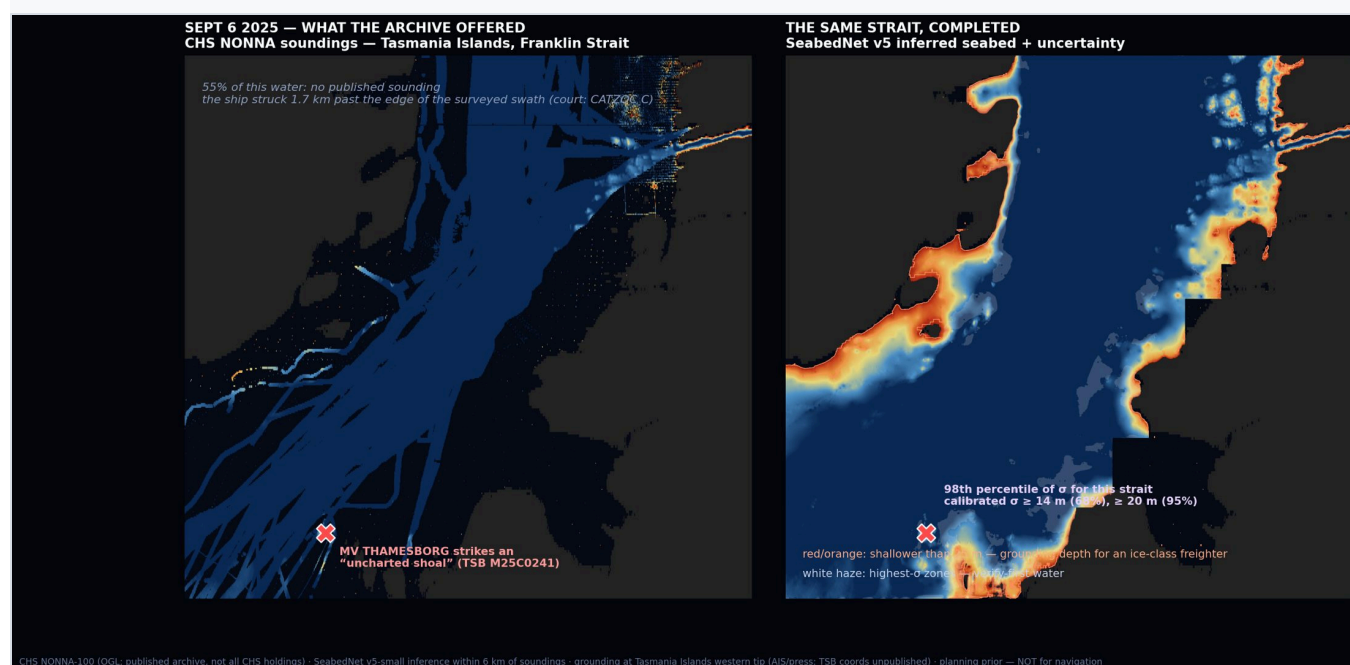


Figure 2. Franklin Strait, 6 September 2025. Left: published soundings (blue), the surveyed swath the court identified as the safer lane, and the strike position. Right: the model-completed depth with its uncertainty; the site sits at the 98th percentile of calibrated σ for the strait (≥ 14 m at 68%, ≥ 20 m at 95%). The nearest dated CHS survey is a 1960 single-beam line, CATZOC C.

6. The corridor product

Completion. 82,121 km² with published soundings → +351,305 km² inferred within 6 km of data (corridor); 429,682 → +1,049,018 km² nationally.

Route provenance. 1150 points at 2 km along the Churchill–Atlantic route: grade A (published sounding) 17%, B (inferred, ≤ 1 km) 27%, C (1–6 km) 34%, D (no inference) 22%.

Least-risk channel. Cost = distance × (shallow-water penalty + $\sigma/8 + 13 \times$ no-data); water under 20 m and land impassable. Result: 2,782 km, shallowest 22 m, mean σ 2.7 m, 0% over no-data water. It is 455 km longer than the straight route: about 20 h at 12 kn, ~21 t of fuel and a day of hire per voyage for a Handysize (~US\$25k). It is a map of where the risk sits, not a sailing direction; the survey plan is what removes the detour.

Of the 198 sounded points, 103 (52%) date from before 1980; median survey year 1974.

Survey plan. Ten highest- σ boxes within 35 km of the route. Coverage: 40 km²/day, from a swath of ~3× depth (300–400 m at 100–130 m), 8 kn for 20 h (~300 line-km, ~100 km² raw), 20% overlap and a 50% weather-and-ice downtime factor; stated as an assumption. Day rate: the CCG polar-icebreaker charter of July 2026 (C\$183,000/day; C\$22M / ~120 days) [10] is the upper bound used in Table 7; the lower bound is the CHS Arctic hydrographic contract to Sedna ROV Services (Iqaluit), C\$6.5M over two seasons, 9,000+ line-km in the first (DFO, March 2024): ~C\$720 per line-km; at a 350 m swath with 20% overlap, ~C\$2,600 per km², which prices the same area at about C\$23M. CCGS Amundsen 2021: ~38,700 km² mapped over an ~80-day season (Copernicus ESSD 2024), i.e. several hundred km²/day in transit mapping of deep water; 40 km²/day is conservative for 100–130 m shelf water.

#	LAT	LON	INFERRED KM ²	MEAN Σ (M)	PEAK Σ (M)	SHIP-DAYS	C\$M
1	59.25°N	59.75°W	1,055	52.6	109.7	26.4	4.8
2	59.25°N	60.25°W	740	55.3	116.6	18.5	3.4
3	59.75°N	60.25°W	559	54.7	102.8	14.0	2.6
4	58.75°N	59.25°W	895	26.1	133.8	22.4	4.1
5	59.25°N	59.25°W	724	30.2	78.2	18.1	3.3
6	59.75°N	60.75°W	993	17.1	62.2	24.8	4.5
7	60.75°N	65.25°W	617	23.9	75.0	15.4	2.8
8	60.75°N	64.75°W	783	17.1	84.7	19.6	3.6
9	58.75°N	58.75°W	1,039	12.2	32.3	26.0	4.8
10	57.75°N	56.75°W	1,348	8.7	33.1	33.7	6.2

Table 7. σ -ranked survey plan: 219 ship-days, C\$40.1M — one fundable season against "more than a decade" for full conventional coverage.

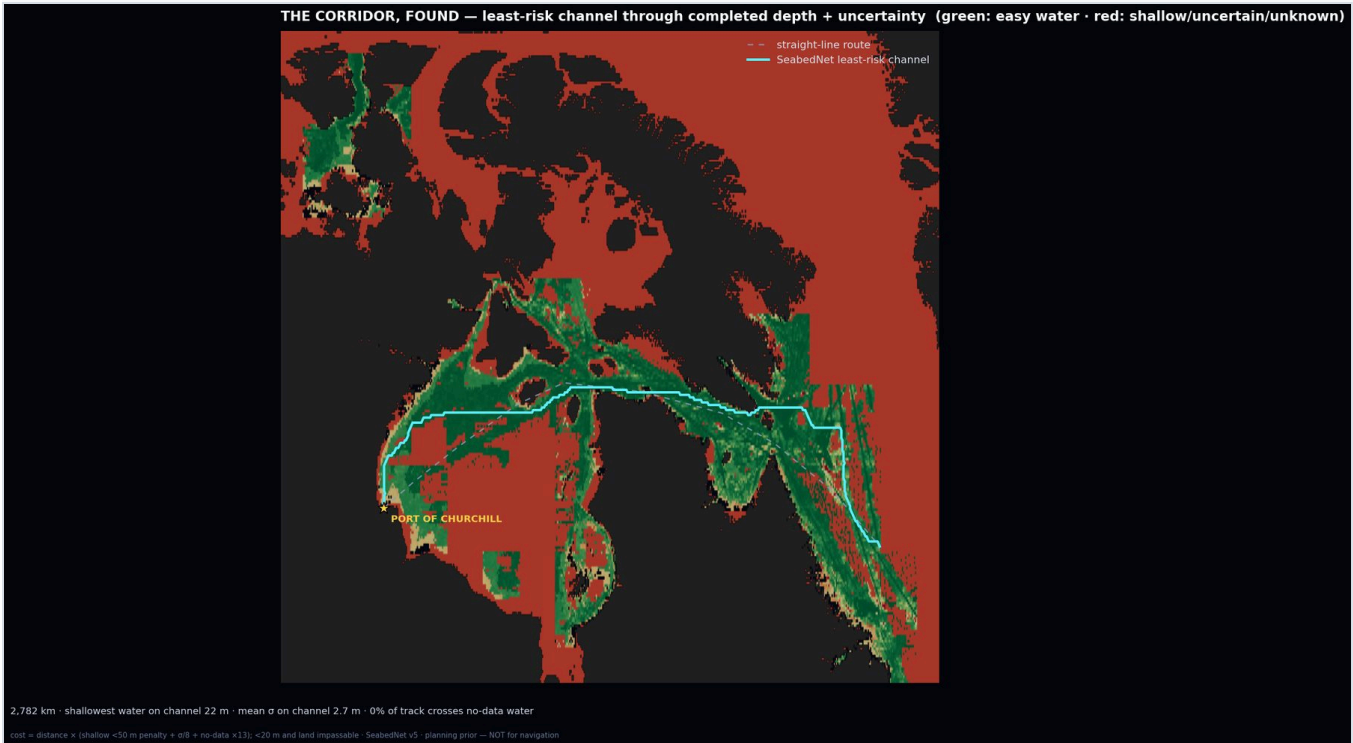


Figure 3. The least-risk channel (cyan) through the completed depth and uncertainty fields; dashed grey is the straight-line route. Green — easy water; red: shallow, uncertain or no-data.

7. A forecast that can fail

File `forecast_2026-09-02.csv` (1,314 cells: 314 on the found channel, the rest random inferred cells across the corridor) lists a predicted depth and calibrated 68% / 95% bands at cells that have no published sounding as of 29 August 2026. SHA-256
4d3f5dc8a5c2b50f3f72fba95384581fc778d6d453a3eec46aa2dc383c66be45, sealed 2026-09-02. Scoring rule: any later survey through these cells reports the mean absolute error and the fraction of soundings inside each band; we publish the score regardless of outcome. The depths were produced by the 34.8M-parameter v5-small completion model with the depth gate and calibrated σ of \$4.3. The file is timestamped with OpenTimestamps (proof `forecast_2026-09-02.csv.ots`, three public calendars, Bitcoin-anchored) and its hash is in a GitHub commit of the validation repository, so the seal does not depend on our clock.

Grading. The forecast does not wait for a dedicated survey. On the first of every month the sealed cells' blocks are re-fetched from the CHS WCS; any sealed cell that has received a published sounding is scored (MAE, bias, band coverage, against nearest-sounding and gravity) and the result is published to the atlas with the per-cell record. The sealed file is hash-checked before every grade.

Benchmark. The temporal split of §3.3 is released as *NONNA-Temporal-Churchill v1*: index rasters for the 94 corridor blocks, the 6,686,037 target cells, the exact WCS request to rebuild the training side from public data, a scorer, and a leaderboard seeded with the baselines of §4.5 and SeabedNet. Anyone can try to beat it; submissions are added as declared.

8. Limitations

(i) NONNA is not the CHS holdings; a "no published sounding" cell may be surveyed. (ii) Test 1 under-represents the glacial shelf; Test 2 covers it but its labels are CHS's own later surveys, so datum and processing differences between survey eras are absorbed into the error. (iii) In water shallower than 50 m the mean-depth model is biased deep by 6–9 m (Table 4); the hazard model, not the mean model, is the appropriate product there. (iv) The hindcast has 7 cases, three of which are controls and two of which lie beyond the publication envelope; the TSB's own count is 74 occurrences for 2000–2018, so this is the public-coordinate subset, a benchmark and not a proof. (v) The Thamesborg position is not yet official. (vi) Survey-day rate and km^2/day are assumptions from public contracts and stated arithmetic, not quotes. (ix) The gravity prior has very likely ingested the Test 1 cruises (§4.5). (vii) The physics prior itself ingests historical soundings, so the gravity baseline is not fully independent on surveyed water; on the cells of Test 1 it is. (viii) Nothing here is a chart. Every product carries "planning prior — not for navigation".

9. Reproducibility

All scripts, result JSON files and the per-cell validation arrays are public at github.com/girardemilio3-svg/seabednet-validation; the atlas page and this report are generated from those files by `build_atlas_v2.py`,

`build_atlas_hazard.py` and `build_report.py`, so no number is transcribed by hand. Data are public (CHS NONNA under the Open Government Licence – Canada; GMRT; CHS Survey Index; SRTM15+). Training the 34.8M completion model to 4,000 steps takes 17 minutes on one RTX 5090 and about 12 hours on an NVIDIA GB10; the hazard model 13 minutes. Model weights are available on request.

10. Discoveries and sealed claims (3 September 2026)

National temporal test. The Test 2 protocol over all NONNA blocks: trained on pre-2016 soundings only, scored on 20,405,461 post-2016 soundings in 300 marine blocks across three oceans: model 12.6 m, nearest sounding 19.7 m, gravity 15.7 m; 74% of errors inside 1σ . On 539,352 cells in inland lakes and fjord heads the gravity anchor is invalid and the model fails with it (79 m vs 47 m nearest): the model is not applicable to inland water.

The Shoal List. The hazard model (§3.2, 34.8M) was run over all 437 blocks. Cells were kept where the completed depth is 21–150 m, $P(\text{shallowest point within 500 m} < 10.5 \text{ m}) \geq 0.8$, the predicted shallowest point lies between the surface and 21 m with relief under 100 m, the cell is within 6 km of a published sounding, no published sounding within ~500 m is shallower than 21 m, and the block's gravity prior agrees with its own soundings to 30 m (excluding inland lakes). Connected clusters of 3–300 cells: 626 nationally. The top 40 by probability, at most two per quarter-degree box, are sealed (`shoal_list_2026-09-03.csv`, SHA-256 `a0bc8c3d2c174491...`, OpenTimestamps). Each is a falsifiable claim scored by a single survey line. Regions: Strait of Belle Isle (7), Beaufort Sea (6), Hecate Strait (5), Chesterfield Inlet (5), Rae Strait / King William Is. (2), Gulf of St. Lawrence (2).

Strike-point predictions. For the four Arctic groundings known only by place name (Nanny 2010, Mokami 2010, Dorsch 2012, Rosaire A. Desgagnés 2025), the hazard model's most dangerous navigable cell in the reported area, clear of charted shore, was sealed before the TSB occurrence positions were requested (`strike_predictions_2026-09-03.csv`, SHA-256 `a33994d6999ab30b...`, OpenTimestamps); hit = within 2 km, graded verbatim on receipt.

Archive audits. Of 215,390,546 sounded cells nationally, 28.0% rest on surveys before 1980 and 8.1% before 1960; 46.8% are post-2016 (Figure 4). Community

resupply lanes (15 km approach zones, 26 communities): the oldest medians are Nauyasat 1955, Deception Bay 1960, Clyde River 1970; Milne Inlet, surveyed for the Baffinland ore trade, is 98% sounded with median 2017 and serves as the control. GEBCO lag: on 73,155 km² of CHS-sounded corridor water shallower than 200 m, the GEBCO-based global grid differs from the archive by more than 10 m on 14.4% of cells and is deep-biased by more than 10 m on 6.0% (4,523 km²); on the route, 15.7% of points differ by more than 10 m and 10.5% are deep-biased (worst 122 m). National σ -ranked survey plan: 39,569 km² in the 20 highest-uncertainty half-degree boxes, 989 ship-days, C\$103–181M.

Negative results. ICESat-2 ATL24 laser bathymetry (SlideRule at 124x, ~3.6M photons over 12 corridor boxes): where the fitted datum offset is physical (± 3 m) the laser agrees with NONNA soundings to 1–5 m, but only 124 photon cells without a published sounding survive the quality gates; the laser cannot score the model's fills in this turbid water. Era audit: post-2016 soundings within 200 m of a dated older sounding differ from it by a median of ~1 m in every decade back to 1900; the age of a sounding does not predict its error, and the risk of old charts lies in the unsounded water between soundings.

THE AGE OF CANADA'S CHART – year of the latest survey under every sounded cell (CHS NONNA + CHS Survey Index)

dark red <1940 · red 1940-60 · orange 1960-80 · amber 1980-2000 · grey-blue 2000-16 · blue post-2016 multibeam | 28.0% of sounded cells predate 1980 · 46.8% are post-2016

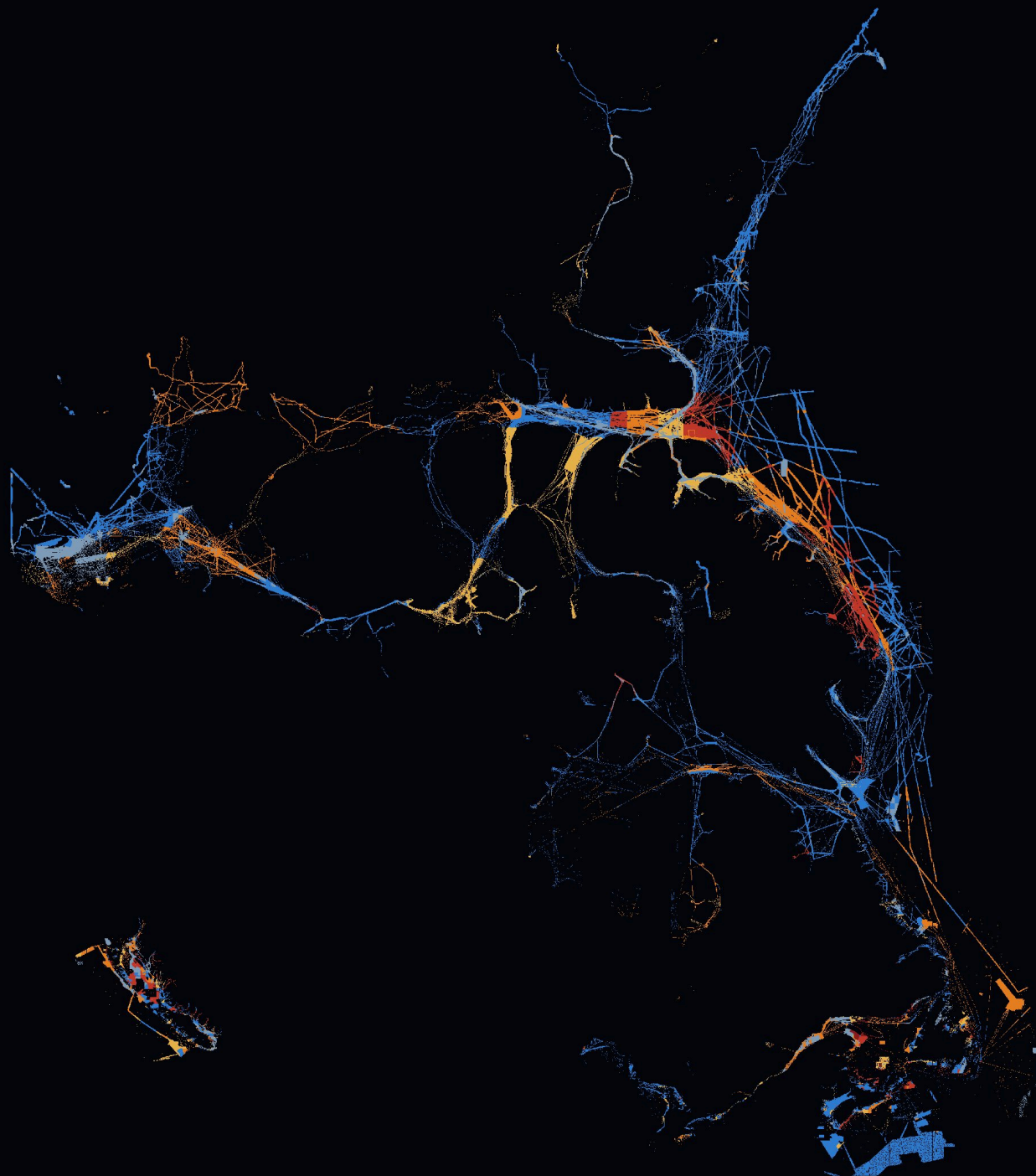


Figure 4. The age of Canada's chart: year of the latest survey under every sounded cell (CHS Survey Index; post-2016 = modern multibeam).

10.1 Correction and Shoal List version 2 (6 September 2026)

Three days after sealing, the version-1 Shoal List was tested against data it had never seen: a Sentinel-2 summer image over every claim, the GSHHG full-

resolution shoreline, and the Coast Guard reported-danger notices of §11. It failed two of our own checks. All forty claims predicted a shallowest point at the surface (0 ± 4 m); 22 of 40 lay within 1 km of a shore; several clustered along single survey tracks (repeated latitudes 58.734° N and 69.397° N). The hazard head had learned that near a coast the shallowest point within 500 m is the beach, which is true and useless, and the top-40-by-probability selection preferentially picked those cells. Version 1 remains sealed and will be scored exactly as published.

Retraining. The shallowest-point head (§3.2) was retrained with two changes to the target: a cell contributes only where at least 95% of its 500 m disc carries 10 m soundings ($\text{COV} \geq 0.95$, which removes coast-adjacent windows and survey gaps; 32% of the original targets survive), and targets at or above -0.5 m are discarded. 4,000 steps, batch 8, seven grounding sites and the geographic holdouts excluded as before. The resulting national field is nearly identical to version 1 away from coasts (per-block correlation 1.00) and differs where it should.

Version 2 filters. To the version-1 criteria: ≥ 2 km from the coastline, predicted shallowest point between 3 and 18 m, cluster of ≥ 5 cells spanning ≥ 2 rows and ≥ 2 columns, > 5 cells from a block edge, $P \geq 0.7$. 5,388 candidate clusters, 272 pass, 40 sealed (`shoal_list_v2_2026-09-06.csv`, SHA-256 958acbdba5938231..., OpenTimestamps). Regions: Queen Maud Gulf (5), Bellot Strait (5), Cape Breton / Sydney Bight (4), Simpson Strait (4), St. Lawrence estuary (3), Ungava Bay (3). Median distance to shore 3.2 km; predicted shallowest points 3.0–6.6 m in 21–52 m water. Sentinel-2 imagery over all forty shows open water with no visible rock or breaker, which at 10 m resolution neither confirms nor refutes a 3–7 m shoal.

11. The reported-dangers test (6 September 2026)

The Canadian Coast Guard publishes every navigational warning in force with a position and a category. All 1,558 positions in the danger categories (Shallow Depth Confirmed, Shallow Depth Reported, Shoal, Uncharted Rock, Submerged Object) from 510 notices in force on 6 September 2026 were parsed; 460 state a measured depth and 94 are Arctic. Each position was scored with the hindcast protocol of §5: the percentile of the hazard field at the position (maximum within 500 m) among apparently-safe cells (mean map deeper than 21 m, within 6 km of a sounding) within 25 km, so that 10% of safe water lies above the 90th percentile by

construction. The nearest-sounding rule (the shallowness of the nearest published sounding, ranked the same way) is the baseline. Notices in the Confirmed category usually follow a CHS survey that may already be in the NONNA snapshot the field was computed from, so the table conditions on the chart still calling the water safe and on the distance to the nearest published sounding.

SUBSET (CHART SAYS SAFE)	N	V1 HEAD, TOP DECILE	V2 HEAD, TOP DECILE	NEAREST- SOUNDING RULE	P VS 10% (V2)
Chart says the water is safe at the position (deeper than 21 m)	107	34 (32%)	36 (34%)	40 (37%)	3e-11
... and no published sounding within 300 m	37	8 (22%)	10 (27%)	7 (19%)	0.003
... and no published sounding within 1 km	22	4 (18%)	3 (14%)	1 (5%)	0.380
Arctic notices only	40	10 (25%)	11 (28%)	9 (22%)	0.001
Notices stating a measured depth of 21 m or less	13	4 (31%)	8 (62%)	5 (38%)	8e-06

Table 8. Reported dangers in the model's top decile of apparently-safe water, by subset. Where a sounding lies beside the notice the simple rule ties or wins; the model's edge appears where no sounding is within 300 m. Samples are small; the full per-notice table is `navwarn/navwarn_hindcast_v2.csv`.

12. The accuracy sprint (10–11 September 2026)

Question. A larger network did not improve the temporal benchmark (§4: 13.5 m tiny vs 13.3 m small), so the limit is information, not capacity. Three new input channels were built for every 100 m block: coastal land elevation from the Copernicus 30 m DEM averaged to the block grid; a Sentinel-2 cloudless summer mosaic (RGB, ~70 m); and a winter (February–April, two seasons) Sentinel-1 backscatter median (VV, VH). A leakage-free gravity anchor was built by masking the ship-constrained cells of SRTM15+ (Scripps SID grid) and refilling from gravity-predicted cells only, with the raw Sandwell V34 gravity anomaly and vertical gradient as channels.

Protocol. Every variant is scored on the corridor temporal benchmark (§4, Test 2: pre-2016 soundings in, post-2016 soundings out, 6,460,612 cells). Fine-tuned variants start from the published checkpoint with the new channels zero-initialised, and are compared with a control fine-tune of identical length and inputs so that extra training is separated from extra inputs. From-scratch variants use the small (6.8M) architecture on the 100 m corpus.

MODEL	MAE MODEL (M)	NEAREST SOUNDING	GRAVITY PRIOR	INSIDE 1 Σ	SHALLOWEST STRATUM
Original (34.8M; soundings + gravity)	13.26	18.85	18.08	74%	15.31
Control fine-tune, same inputs	12.67	18.85	18.08	68%	13.65
Fine-tune + coastal elevation + Sentinel-2	13.18	18.85	18.08	67%	17.97
From scratch (6.8M): soundings + gravity	14.57	18.85	18.08	70%	16.25
From scratch: + coastal elevation + Sentinel-2	14.83	18.85	18.08	66%	22.15
From scratch: + winter Sentinel-1	13.88	18.85	18.08	68%	15.76
From scratch: + raw gravity, leakage-free anchor	15.48	18.85	26.52	69%	20.86

Table 9. Temporal benchmark by variant. Rows appear as runs complete.

Result so far. The control fine-tune improves the published model by 0.6 m; the same fine-tune with coastal elevation and Sentinel-2 added is worse (13.18 m) and worse still in the shallowest stratum. Fine-tuning a converged network with zero-initialised inputs is a weak test of the inputs; the from-scratch rows are the decisive ones.

The archived reported dangers. The Coast Guard’s cancelled navigational warnings add 8,948 danger positions (2017–2026; 3,302 with a stated depth; 385 Arctic) to the 1,558 in force (§11). Scored with the §11 protocol (Table 10), the

hazard head and the nearest-sounding rule tie where the chart calls the water safe, and the rule wins in the Arctic and on notices with a measured depth. At this sample size the hazard head, as published, does not beat the obvious heuristic.

SUBSET (ARCHIVE, CHART SAYS SAFE)	N	HAZARD HEAD IN TOP DECILE	NEAREST-SOUNDING RULE
chart says safe	1373	682 (50%)	662 (48%)
+ no sounding within 300 m (blind)	124	39 (31%)	37 (30%)
+ no sounding within 1 km	66	11 (17%)	9 (14%)
+ nearest sounding deeper than 21 m	1298	644 (50%)	603 (46%)
Arctic, chart safe	119	28 (24%)	56 (47%)
stated depth <=21 m, chart safe	129	61 (47%)	91 (71%)
Reported/Shoal/Uncharted Rock only, chart safe	285	116 (41%)	157 (55%)

Table 10. Archived reported dangers in the top decile of apparently-safe water within 25 km, version-2 hazard head vs the nearest-sounding baseline.

A model of where dangers are reported. Every dated notice in chart-safe water (live and archived) against ten random chart-safe cells per notice from the same blocks; features: hazard probability and predicted shallowest point, completed depth and σ , nearest published sounding depth and distance, coastal elevation at the cell and its 500 m maximum, Sentinel-2 RGB, winter Sentinel-1 VV/VH, gravity prior, latitude, distance to the coast. Gradient-boosted trees trained on notices to 2022 and tested on 2023–2026 (39,050 cells, 692 reported dangers, base rate 1.8%).

RANKING	AUC	TOP-DECILE PRECISION	TOP-DECILE RECALL
All cues	0.932	14.6%	82%
All cues except the hazard field	0.931	14.4%	81%
Hazard field only (learned)	0.857	10.7%	61%
Nearest sounding only (learned)	0.846	9.9%	56%
Hazard probability, raw rank	0.741	3.0%	17%

RANKING	AUC	TOP-DECILE PRECISION	TOP-DECILE RECALL
Nearest sounding depth, raw rank	0.724	2.8%	16%

Table 11. Danger-report model on years it never saw. Drop-one AUC loss, largest first: nearest_km (+0.015), coast_dist_px (+0.008), shoal_m (+0.004), land_max500 (+0.004), s2_r (+0.003), s1_vv (+0.003).

One tenth of the chart-safe water captures 82% of the later reported dangers, 8× the base rate. The hazard field contributes almost nothing once the other cues are present, and the two strongest cues, distance to the nearest sounding and to the coast, are in part measures of where ships go. This is therefore a model of reporting as much as of the seabed: the correct target for deciding where a survey launch goes, and not a statement about bathymetry.

Survey plans, re-ranked. §7 ranked boxes by uncertainty. Two operational rankings are added: by expected keel-depth hazard area per ship-day from the hazard field (1,972 km², 49 ship-days, C\$5–9M for the top twenty), and by expected reported-danger area per ship-day from the model above (4,973 km², 124 ship-days, C\$13–23M), both restricted to water within 30 km of NRCan’s Arctic shipping routes or the Churchill route or 15 km of a sealift community, inland lakes excluded.

BOX	AREA (KM ²)	EXPECTED REPORTED-DANGER AREA (KM ²)	SHIP-DAYS	PER SHIP-DAY
64.0°N 83.5°W	118	26.7	3	9.08
64.5°N 78.5°W	55	8.0	1	5.77
69.0°N 94.0°W	31	4.3	1	5.51
63.5°N 92.0°W	58	6.8	2	4.70
62.5°N 65.5°W	138	14.6	3	4.25
63.5°N 69.0°W	175	16.5	4	3.77

BOX	AREA (KM ²)	EXPECTED REPORTED-DANGER AREA (KM ²)	SHIP- DAYS	PER SHIP- DAY
63.0°N 91.0°W	289	24.1	7	3.33
65.5°N 59.0°W	90	6.1	2	2.73
63.5°N 91.5°W	143	9.4	4	2.61
62.5°N 91.5°W	580	37.5	14	2.58

Table 12. Top ten boxes by the danger-report model (file national_plan_v3.json; hazard-field ranking in national_plan_v2.json).

Second pass. Adding the log(blue/green) satellite-bathymetry index, radar texture (std of VV within 500 m), seabed roughness and slope of the completed depth, the distance to the nearest shallow (< 10 m) sounding, and the mean hazard within 1 km raises the combined ranking to AUC 0.942 (83% recall in the top decile).

Arctic-only test (1,377 cells, 105 reported dangers, base rate 7.6%): top-decile precision 45%, 6x the base rate, recall 58%. Removing every exposure cue (distances to soundings, to shallow soundings, to the coast, latitude) leaves AUC 0.909 and 77% recall. The top decile is predicted at 10% and observed at 15% (isotonic reliability on the test years). Files danger_model2.json, danger_model2.py.

Measured depths at reported shoals. 3,488 notices state a least depth (Table 13). Where the chart called the water safe (n = 197, reported median 14 m), the predicted shallowest point is unbiased (-1.1 m; nearest sounding +19 m) but not more precise (MAE 19 m vs 19 m; within 5 m in 34% vs 31% of cases). Arctic (n = 291): model 10.9 m, nearest sounding 8.6 m, chart bias +6.7 m. File navwarn_depth_test.json.

13. Ensemble, blend, ships of opportunity and the first prospective score (12–19 September 2026)

Test-time augmentation and ensembling. Averaging the full-size control model over its four flips at inference improves the temporal benchmark from 13.18 to 12.89 m at no training cost (Table 14). An equal-weight ensemble of the published model and two full-data retrains scores 12.83 m. Nearest-sounding copying beats every model inside about 500 m of a sounding (Table 3), so a distance-weighted blend, $est = w(d) \cdot model + (1 - w(d)) \cdot nearest$, was fitted by least squares on the odd-numbered corridor blocks and scored on the even-numbered ones: 12.45 m against 12.68 m for the ensemble alone. The fitted weights (model share, mean of the two folds; 0 km: 0.00, 0.25 km: 0.38, 0.5 km: 0.73, 0.75 km: 0.84, 1.0 km: 0.89, 1.5 km: 0.92, 2.0 km: 0.93, 3.0 km: 0.90) are near zero inside 250 m and above 0.85 beyond 1 km. The published corridor map is now a six-member ensemble (three seeds added) with this blend; its members saw every year of soundings, so the three-member blended figure is its held-out proxy.

MODEL	MAE (M)	0–20 M	20–50 M	INSIDE 1Σ
Full-size control, no augmentation	13.18	14.60	14.89	69%
Full-size control, 4-fold test-time augmentation	12.89	13.99	14.44	71%
Three-member ensemble	12.83	13.60	15.20	72%
Three-member augmented ensemble + distance blend (weights fitted on half the blocks, scored on the other half)	12.45	12.90	14.88	74%

Table 14. Temporal benchmark (pre-2016 training, post-2016 corridor soundings) for augmentation, ensembling and the distance blend.

Ships of opportunity as an independent test. The IHO crowdsourced bathymetry archive (NOAA NCEI DCDB) was crawled for every ping in Canadian waters from 2017 to 2026 (about 450 million). Vessels were calibrated against charted soundings and kept only if at least 300 of their pings on sounded cells agreed with the chart to a median absolute deviation of 3 m or less, with the vessel's median offset removed (44 of 82 vessels; residual 1.6 m median on sounded cells). Cells reporting under 20 m where a survey within 100 m reads over 100 m were dropped as sounder dropouts. The remaining 278,898 shelf cells (0–200 m) with

no charted sounding, almost all on the Pacific and Atlantic coasts, give Table 15. The pool holds no Arctic pings: sealift and expedition operators do not contribute to it.

CELLS	N	MODEL MEDIAN ERROR (M)	WITHIN 10 M	NEAREST SOUNDING MEDIAN (M)	WITHIN 10 M
All shelf cells, three-member map	278898	4.9	70%	3.2	83%
0-0.25 km from a sounding	133196	4.3	75%	2.5	94%
0.25-0.5 km from a sounding	68247	4.7	71%	3.0	91%
0.5-1 km from a sounding	38776	5.5	64%	3.8	85%
1-3 km from a sounding	13774	6.3	64%	8.3	56%
3-6 km from a sounding	6715	8.1	56%	25.3	34%
All shelf cells, six-member blended map	291818	3.4	81%	3.2	83%

Table 15. Ship soundings nobody trained on, at map cells with no charted sounding. The model is worse than the nearest charted sounding inside a kilometre and better beyond it; the blended six-member map closes the gap (last row).

Ships against the claims. The same pings were run against every sealed shoal claim and every hazard candidate. One sealed claim is refuted: St. Lawrence estuary at 47.633°N 70.046°W, predicted 3 m where the chart reads about 30 m, has 613 ship soundings within 300 m reading 27.2–36.7 m. The other 39 sealed claims are in Arctic water without ship traffic. Of 15,936 unsealed hazard candidates, 217 have at least five pings within 300 m: 196 refuted, 16 ambiguous, none confirmed as an offshore shoal once one vessel logging a constant 0.7 m and one coastal-shallows site are removed. The hazard field over-predicts in charted,

busy southern water; because vessels choose safe water, a sounding 300 m away does not exclude a pinnacle narrower than 100 m.

Laser altimetry, second pass. NASA's ATL24 bathymetry product returns no photons at any of the claim sites (its search mask excludes them; a Bahamas control returns thousands), so the raw ATL03 photons over all 36 claims with data were re-analysed with an along-track detector: local sea surface per 20 m bin, densest 0.5 m depth slice 2.5–40 m below it per 10 m bin with a Poisson test against the pass's own 15–40 m noise, a surface-to-bottom gap requirement, at least six consecutive bins within 1.5 m, and a guard discarding flat segments 3.3–5.0 m below the surface (the detector afterpulse). On 69 passes over two deep-water control sites (charted 300–400 m) it found no bottom. Over the claims it found a bottom on 1 of 36 sites, on one pass each, never on two. The earlier apparent detections were the surface tail. In ice-affected Arctic water the laser cannot see 3–7 m shoals often enough to confirm or refute them.

Prospective test, first reading. Two danger-report forecasts were sealed before any outcome: the Arctic and Hudson Bay corridors on 2026-09-12 (8,851,961 cells, SHA-256 d57eb0fb858e..., OpenTimestamps proof) and the southern coasts on 2026-09-13 (4,634,780 cells, afe70e97fbdf...). The rule, fixed at sealing: a Coast Guard danger notice issued after the seal with a position within 1 km of a scored cell counts, and a hit is a notice whose nearest cell is in the national top decile (chance 10%) or top percentile (chance 1%), scored per message by its best position. As of 2026-09-18 the Arctic file has no eligible notice. The southern file has 10 messages inside it; 9 are in the top decile and 9 in the top percentile (Wilson 95% interval for the decile share 60–98%). Nine of the ten are one survey campaign in Chedabucto Bay, Nova Scotia, issued four days after the seal placed the bay in the top 1% nationally; the bay produced 132 danger notices from 2019 to 2025, all in the model's training set, so this is a hit on a recurring area, not the discovery of an unknown one. The tenth, in the St. Lawrence estuary, sits at the 47th percentile. Two independent events, one hit and one miss; the interval treats notices as independent and overstates the evidence. The files are re-scored weekly.

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Version 1.3, 20 September 2026 (§13 added). Generated from result files;
see the repository for the exact commit.